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# Anthropic Models on Amazon Bedrock

Module 3:

Data Preparation and Knowledge Bases for RAG

Lab 1

# Course Contents

Module 1: Foundations: Generative AI and Claude Models on AWS

Module 2: Claude API Fundamentals and Prompt Engineering

▸ **Module 3: Data Preparation and Knowledge Bases for RAG**

Module 4: Tool Use and Agentic Workflows with Claude

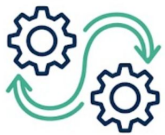
Module 5: Security, Governance, and Guardrails

Module 6: Practical Architecture and Deployment Patterns

Module 7: Claude Code: Building Bedrock Applications

Module 8: Amazon Kiro and Enterprise Code Generation

# Module Objectives



## RAG Architecture

Understand RAG benefits for enterprise applications

## Knowledge Bases

Set up Bedrock Knowledge Bases with data sources

## Retrieval Queries

Query knowledge bases with citation support

## Optimize RAG Performance

Apply reranking and chunking strategies

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# RAG Architecture and Concepts

# Knowledge Bases for Amazon Bedrock

Managed RAG for Amazon Bedrock

Connect your data to foundation models without code

Automatic document chunking, embedding, and retrieval

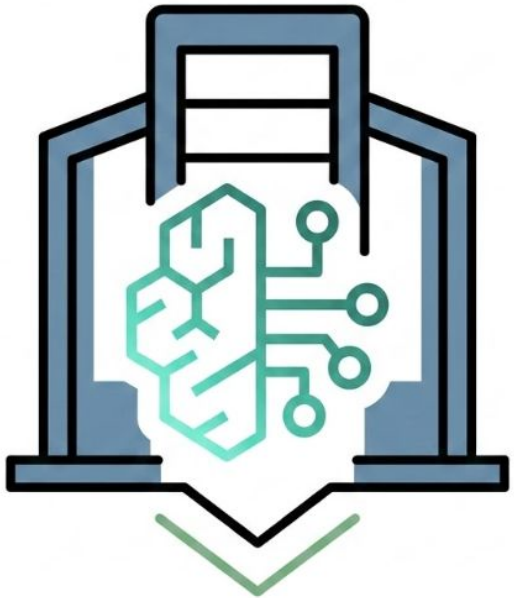
Fully managed vector store with multiple engines

Query via API or use directly in your application

# The Enterprise AI Challenge

## The Gap

- Claude doesn’t know your internal policies
- Proprietary docs aren’t in training data
- Fine-tuning is expensive, needs retraining
- Context windows have token/cost limits
- Solution: Give Claude what it needs, when needed



# What is RAG?

Retrieval-Augmented Generation

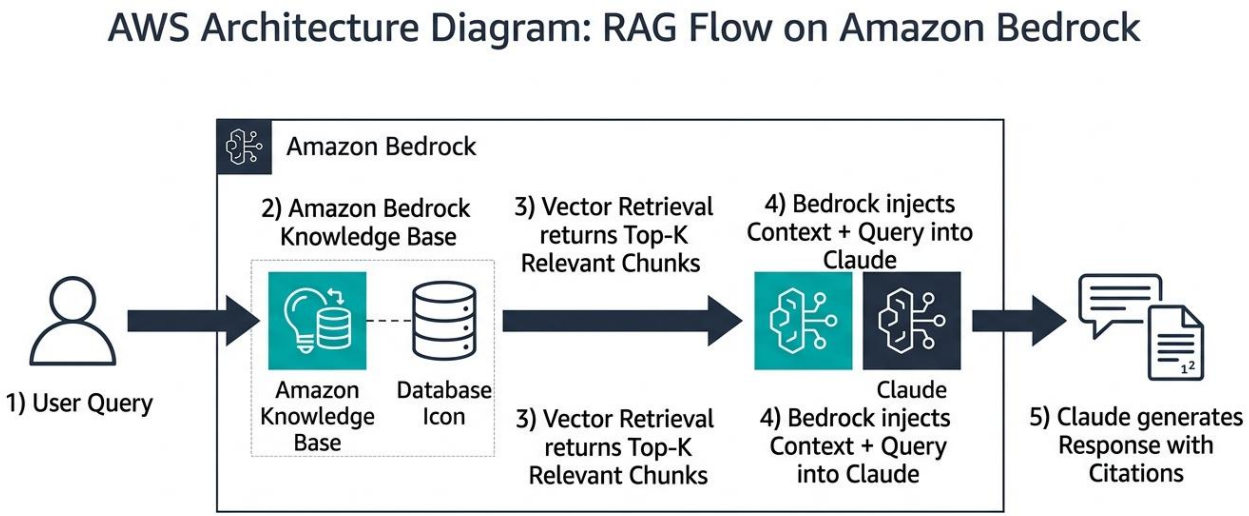
Retrieval: Find relevant info from your data

Augmented: Add that info to prompt context

Generation: Claude responds using retrieved data

Strategic context use — like a librarian, not a memorized encyclopedia

# RAG Architecture Overview



# Why RAG for Enterprise?

## Accuracy

Answers grounded in your documents

## Currency

Updates without retraining the model

## Control

You decide what the model accesses

## Cost

Only retrieve what's needed

# Vector Embeddings Explained

## Embeddings 101

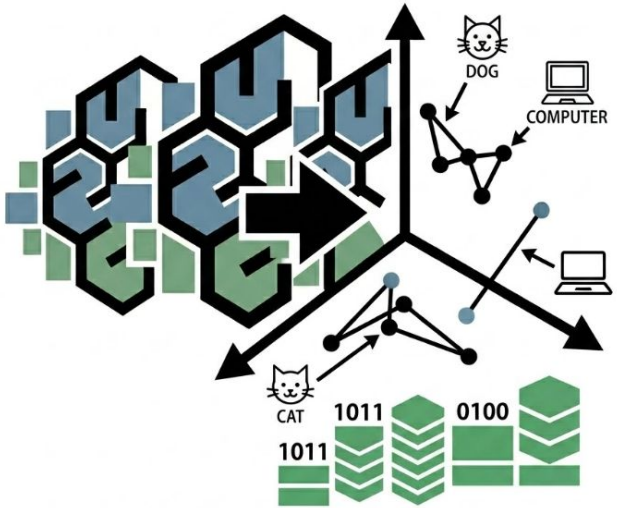
Text converted to numerical vectors

Similar meanings produce similar vectors

'Return policy' close to 'refund process'

Retrieval = finding vectors closest to query

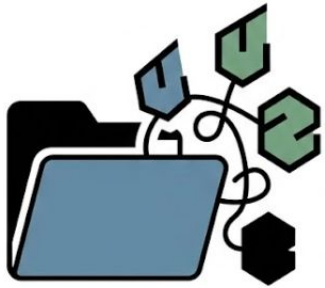
Semantic search: by meaning, not keywords



# Knowledge Bases for Amazon Bedrock

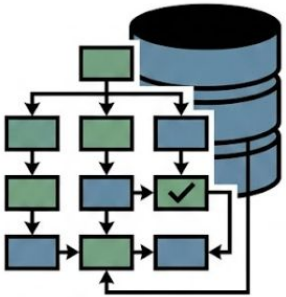
1. **Data Sources:** S3, web crawlers, structured databases
2. **Embeddings:** Auto-conversion via Titan or other models
3. **Vector Store:** OpenSearch Serverless or bring your own
4. **Retrieval API:** Query interface with citation support
5. **Fully Managed:** No infrastructure to provision

# Supported Data Types



## Unstructured

PDFs, Word docs
Text, HTML, and
Markdown files

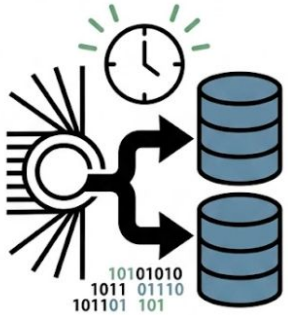


## Structured

Database
connections
with natural language
to SQL queries

## Multimodal

Docs with images
plus image-based
content search




## Real-time

Immediate ingestion
for live data sources
as they update

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# Configuring Amazon Bedrock Knowledge Bases

# Knowledge Base Configuration Steps

1. **Name and description** — Be descriptive for multiple KBs
2. **IAM role** — Permissions to read from data sources
3. **Data source** — S3 bucket, prefix, sync schedule
4. **Embedding model** — Amazon Titan, Cohere, or custom
5. **Vector store** — OpenSearch Serverless or your own

# Chunking Strategies

| Default | Fixed-Size | Semantic |
| --- | --- | --- |
| Bedrock decides how to split documents for most workloads<br>No configuration needed | You specify the chunk size in tokens for precise control over retrieval granularity | Chunks split at natural content boundaries like paragraphs and sections for best meaning fit |

# Embedding Model Options

| Model | Best For | Notes |
|---|---|---|
| Amazon Titan Embeddings V2 | Most use cases | Default, works well for English |
| Titan Multimodal Embeddings | Documents with images | Search across text and visuals |
| Cohere Embed | Multilingual content | Strong non-English support |
| Custom models | Special requirements | Bring your own if needed |

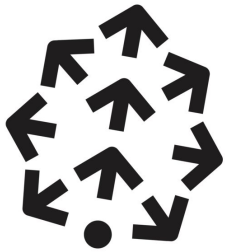
# Vector Store Options

**OpenSearch**

Serverless and fully managed by AWS
Recommended default

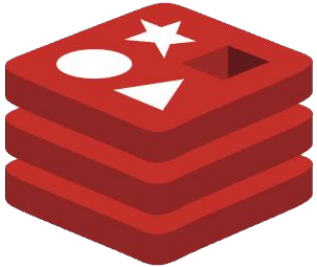
**Aurora pgvector**

PostgreSQL-based vector storage with relational features



**Pinecone**

Hosted vector DB for cross-app sharing and existing setups



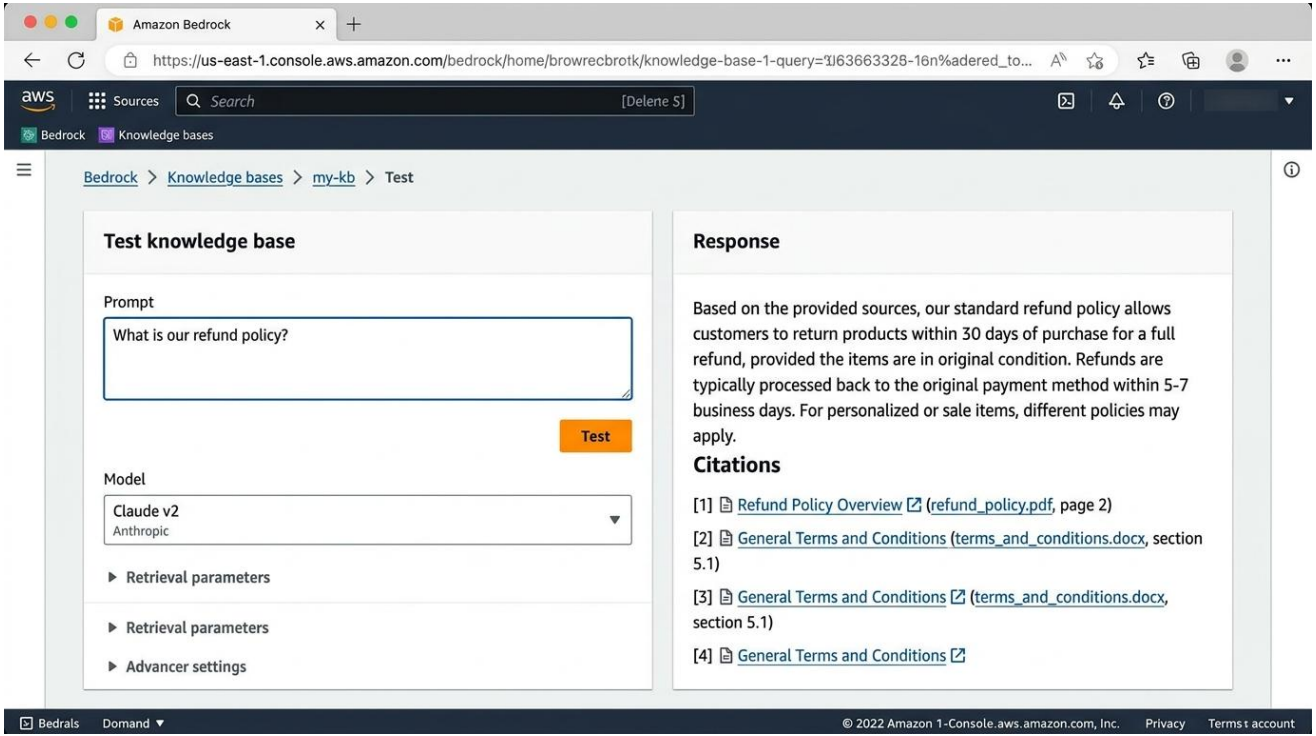
**Redis**

In-memory vector store for low-latency retrieval workloads

# Data Sync Options

1. **Manual sync:** Trigger on-demand from console or API
2. **Automatic sync:** Schedule regular updates (hourly, daily)
3. **Real-time ingestion:** Documents available immediately
4. **Incremental updates:** Only process changed files
5. **Plan for sync time** — not instantaneous in app design

# Testing in the Console



# Demo

- Create a knowledge base in the Bedrock console
- Configure S3 data source and sync documents
- Test RAG queries with natural language questions
- Review retrieved chunks, scores, and citations

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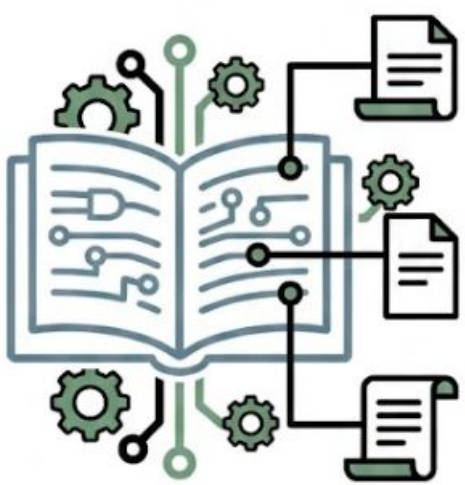
# Optimization and Best Practices

# Citation Support for Source Verification

## Source Attribution

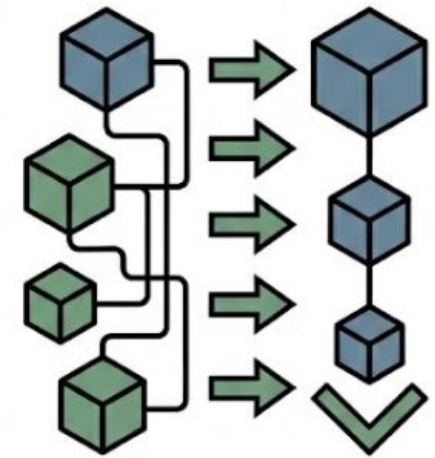
- Automatic source attribution per response
- Document name, page, chunk location
- Users verify by clicking through citations
- Critical for compliance and regulated industries
- Enabled by default in RetrieveAndGenerate API



# Reranking for Better Results

## Better Retrieval

- Initial retrieval: fast vector similarity
- Reranking: second-pass relevance scoring
- Models: Amazon Rerank, Cohere Rerank
- Trade-off: 100-300ms additional latency
- Worth it for most applications



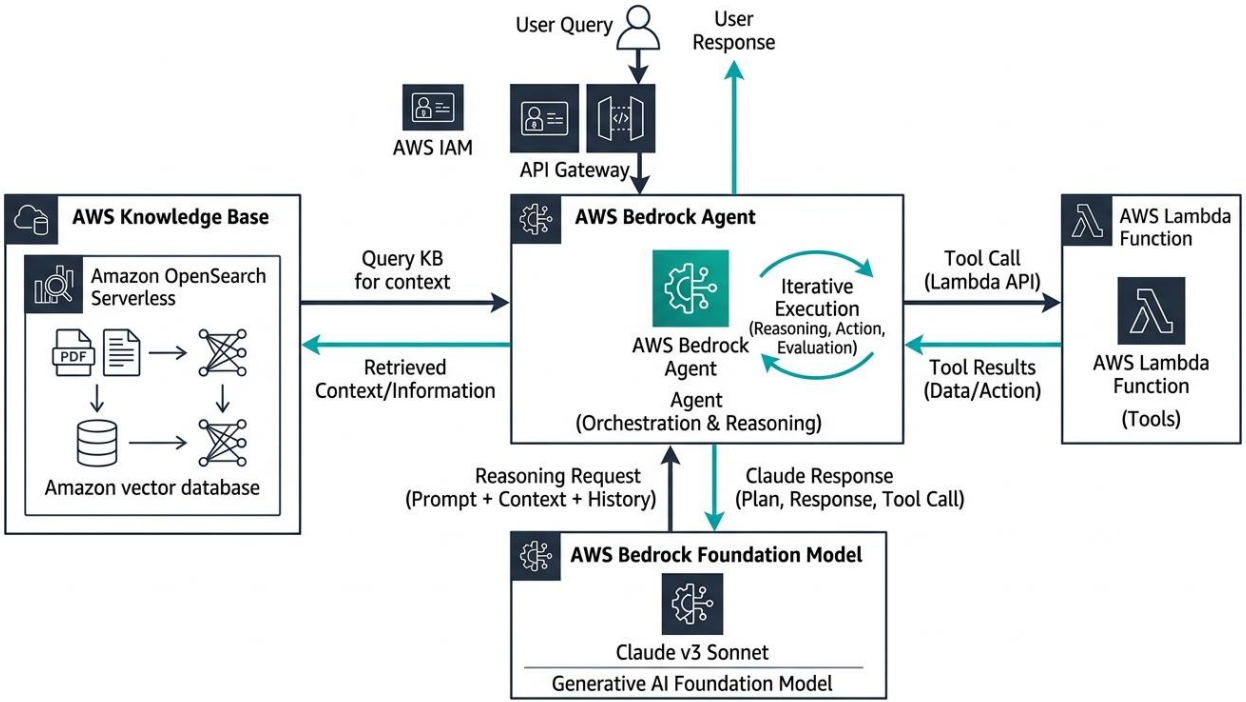
# Chunking Best Practices

1. **Too small:** Context is fragmented, meaning is lost
2. **Too large:** Retrieval is imprecise, context window fills up
3. **Sweet spot:** 200-500 tokens with overlap
4. **Overlap:** Ensures important context isn't split
5. **Semantic chunking:** Break at natural boundaries when possible

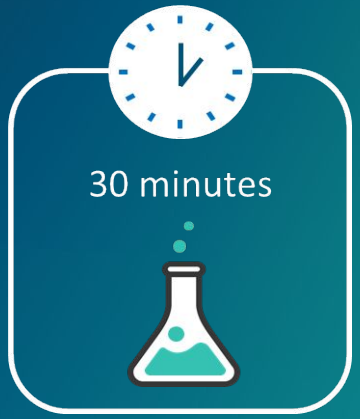
# Query Optimization Techniques

1. **Reformulation:** Rewrite vague queries first
2. **Metadata filtering:** Narrow by source or date
3. **Hybrid search:** Vectors plus keyword matching
4. **Result count:** Balance relevance vs. completeness
5. **Iterate:** Test with real user questions early

# Knowledge Base + Agent Integration



# Lab 1

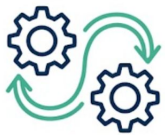



## Claude on Amazon Bedrock with RAG

In this lab, you will:

- Differentiate and invoke Claude Models on Amazon Bedrock
- Observe latency and token usage
- Create a knowledge base from sample documents
- Verify accuracy of RAG citations

# What We Covered



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